

Xiuying Wang

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EDUCATION

- **Queen Mary University of London** Sep 2023 – July 2027
Bachelor of Science (Engineering)
- **Beijing University of Posts and Telecommunications** Sep 2023 – July 2027
Bachelor of Science (Engineering)
Average Mark: 92; GPA: 3.81/4.0 (Ranking Top 10%)

PUBLICATIONS

- [1] **ICML 2026** Lightweight Federated Incremental Learning via Decoupled Replay.
Xiuying Wang, Yichen Li, Hang Su, Gaozhao Liu, Shiwei Li, Chuang Zhao, Jiangming Shi, Imran Razzak
- [2] **TPAMI** Ego-R1: Chain-of-Tool-Thought for Ultra-Long Egocentric Video Reasoning.
S Tian, R Wang, H Guo, P Wu, Y Dong, Xiuying Wang, Jingkang Yang, Hao Zhang, Hongyuan Zhu, Ziwei Liu
- [3] **NeurIPS 2025** Feature Distillation is the Better Choice for Model-Heterogeneous Federated Learning.
Yichen Li, Xiuying Wang, Wenchao Xu, Haozhao Wang, Yining Qi, Jiahua Dong, Ruixuan Li

UNDER REVIEW

- [4] **NeurIPS 2026** Rehearsal-Free Statistical Prototype Regularization for Federated Incremental Learning
Xiuying Wang, Yichen Li, Jiahua Cheng, Xiwei Liu, Chuang Zhao, Bo Liu, Imran Razzak
- [5] **NeurIPS 2026** Heterogeneity-aware Distillation for Federated Continual Learning.
Gaozhao Liu, Yichen Li, Xiuying Wang, Yulong Li, Chuang Zhao, Yankai Jiang, Imran Razzak

RESEARCH EXPERIENCE

- **Multimedia Lab, Nanyang Technological University (MMLab-NTU)** Oct 2024 – Mar 2026
Working closely with Jingkang Yang (Ph.D.) in Ziwei Liu's team Singapore
 - EgoLife: contributed to the EgoLife team by supporting the construction and annotation of the EgoLife dataset, and helped develop EgoRAG, a retrieval-augmented generation system for long-form egocentric video understanding.
 - Ego-R1 [2]: participated in developing a reasoning-agent framework and benchmark for long egocentric video understanding, focusing on tool-augmented reasoning and evaluation design.
 - Visual Generation: surveyed benchmarks for visual generation tasks, summarizing recent advances, representative methods, evaluation protocols, and emerging research directions.
- **Intelligent and Distributed Computing Lab, HUST & GenMI Research Lab, MBZUAI** Mar 2024 – Present
Working closely with Yichen Li (Ph.D. Student) Wuhan, China
 - Heterogeneous Federated Learning [3]: developed a feature-distillation framework to address model heterogeneity in federated learning.
 - Federated Continual Learning: led research on frameworks that leverage lightweight features for decoupled replay [1] and statistical prototypes for regularization [4], mitigating catastrophic forgetting in distributed settings.

WORKING EXPERIENCE

- **Synvo.ai & MMLab-NTU** Aug 2025 – Mar 2026
Research Engineer Intern in Chen Change Loy's team Singapore
 - AI Copilot for Retail Scenarios: led the end-to-end development pipeline, including model training, agent construction, deployment, encryption, and system optimization, to build a VLM-powered agent system for retail stores with detection, segmentation, and ReID capabilities.
 - PoC for a Smart Glass Company: contributed to the construction of long-horizon memory for egocentric streaming videos captured by always-on smart glasses in everyday scenarios.
 - Security for VLMs: conducted research on feature inversion attacks against VLMs in egocentric scenarios, analyzing privacy risks and investigating potential attack pipelines and defense mechanisms.
- **Action Intelligence** May 2026 – Present
Research Engineer Intern in Jingkang Yang's team Shanghai, China
 - Scalable Egocentric Data Collection for Embodied AI: designing and implementing a multimodal data-collection pipeline that integrates smart glasses, IMU sensors, AprilTag-based calibration, hand-pose estimation algorithms, and customized hardware to collect scalable egocentric interaction data for embodied intelligence research.

RESEARCH INTERESTS

- **Egocentric Intelligence:** Egocentric video understanding, long-horizon memory, multimodal data collection, and embodied interaction datasets.
- **VLM Systems:** Vision-language models, agentic multimodal systems, RAG, and real-world deployment for interactive AI applications.
- **Federated Learning & Security:** Model-heterogeneous federated learning, privacy-preserving learning, and security analysis.
- **Continual Learning:** Federated continual learning and catastrophic forgetting mitigation.

SKILLS

- **Programming:** Python, C/C++, Java, MATLAB
- **Deep Learning & Multimodal AI:** PyTorch, Transformers, LLMs/VLMs, RAG, agentic multimodal systems
- **Computer Vision & Egocentric Systems:** Detection, segmentation, ReID, hand-pose estimation, AprilTag calibration, smart glasses/IMU data collection
- **Machine Learning:** Federated learning and continual learning
- **Tools:** Model deployment, distributed training, vector databases, Linux, Git, Docker

LANGUAGE

- **English:** CEFR B2
- **Chinese:** Native